

Deterministic Optimization

Linear Optimization Modeling

Decision-Making under Uncertainty

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Two-Stage Stochastic Linear
Programming

Modeling using Linear Programs

Learning Objectives for this Lesson

- Discover a Two-Stage Stochastic Programming Model for Inventory Control
- Discover reformulations for discrete random variables
- Discover general two-stage stochastic programs

Inventory Control Problem

- A company designs, produces, and sells winter fashion items such as ski jackets.
- The company must commit to specific production quantity x before knowing the exact demand d , 3 months before the winter season.
- Demand d is estimated as a random variable.
- After seeing demand, the company decides how many to sell and how many to sell at a discounted price s .
- This is called *Decision Making under Uncertainty*, because decision x is made under uncertain demand d .



Stochastic Inventory Control

- Decision variables:
 - **Here-and-Now decision:** production quantity x
 - **Wait-and-See decision:** Sell quantity y , Discount (savage) quantity z
- Objective: minimize production cost and expected future cost
- Stochastic program:

$$\begin{aligned} & \min_x cx + E_d[Q(x, d)] \\ & \text{s.t. } 0 \leq x \leq \bar{x} \end{aligned}$$

First-Stage Problem with q as first-stage decision

$$\begin{aligned} Q(x, d) = & \min_{y, z} -r \cdot y - s \cdot z \\ & \text{s.t. } y \leq d \\ & \quad y + z \leq x \\ & \quad y \geq 0, z \geq 0 \end{aligned}$$

Second-Stage Problem with second-stage decision y, v

Second-stage decisions depend on d
i.e. y and z are functions of d

Reformulations

$$\begin{aligned} \min_x \quad & cx + E_d[Q(x, d)] \\ \text{s.t.} \quad & 0 \leq x \leq \bar{x} \end{aligned}$$

$$\begin{aligned} Q(x, d) = \min_{y, z} \quad & -r \cdot y - s \cdot z \\ \text{s.t.} \quad & y \leq d \\ & y + z \leq x \\ & y \geq 0, z \geq 0 \end{aligned}$$

$$\begin{aligned} \min_x \quad & cx + E_d \left[\min_{y(d), z(d)} -ry(d) - sz(d) \right] \\ \text{s.t.} \quad & 0 \leq x \leq \bar{x} \\ & y(d) \leq d \quad \forall d \in D \\ & -x + y(d) + z(d) \leq 0 \quad \forall d \in D \\ & y(d) \geq 0, z(d) \geq 0 \quad \forall d \in D \end{aligned}$$

$$\begin{aligned} \min_{x, y(\cdot), z(\cdot)} \quad & cx + E_d[-ry(d) - sz(d)] \\ \text{s.t.} \quad & 0 \leq x \leq \bar{x} \\ & y(d) \leq d \quad \forall d \in D \\ & -x + y(d) + z(d) \leq 0 \quad \forall d \in D \\ & y(d) \geq 0, z(d) \geq 0 \quad \forall d \in D \end{aligned}$$

$y(\cdot)$ and $z(\cdot)$ are sales **policies**:

Given a demand d , gives quantities $y(d)$ and $z(d)$

Discrete Random Variable

- Suppose demand d is a discrete random variable with S scenarios, $d_1, \dots, d_S$, each d_i with a probability p_i .
- Policy $y(\cdot)$ has values $y_1, \dots, y_S$, corresponding to d_i 's.

$$\begin{aligned} & \min_{x, y(\cdot), z(\cdot)} cx + E_d[-ry(d) - sz(d)] \\ \text{s.t.} \quad & 0 \leq x \leq \bar{x} \\ & y(d) \leq d \quad \forall d \in D \\ & -x + y(d) + z(d) \leq 0 \quad \forall d \in D \\ & y(d) \geq 0, z(d) \geq 0 \quad \forall d \in D \end{aligned}$$



$$\begin{aligned} & \min_{x, y_i, z_i} cx + \sum_{i=1}^S p_i(-ry_i - sz_i) \\ \text{s.t.} \quad & 0 \leq x \leq \bar{x} \\ & y_i \leq d_i \quad \forall i = 1, \dots, S \\ & -x + y_i + z_i \leq 0 \quad \forall i = 1, \dots, S \\ & y_i \geq 0, z_i \geq 0 \quad \forall i = 1, \dots, S \end{aligned}$$

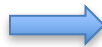
A deterministic LP

General Two-Stage Stochastic Program

- A general two-stage stochastic program can be formulated as

General random variable

$$\begin{aligned} & \min_{x, y(\cdot)} cx + E_w[q(w)^\top y(w)] \\ \text{s.t. } & Ax \leq b \\ & T(w)x + Wy(w) \leq h(w) \quad \forall w \in W \\ & y(w) \geq 0 \quad \forall w \in W \end{aligned}$$



Discrete random variable

$$\begin{aligned} & \min_{x, y_i} cx + \sum_i^S p_i q_i^\top y_i \\ \text{s.t. } & Ax \leq b \\ & T_i x + W y_i \leq h_i \quad \forall i = 1, \dots, S \\ & y_i \geq 0 \quad \forall i = 1, \dots, S \end{aligned}$$

q_i, T_i, W_i, h_i are scenarios of discrete r.v. $q(w), T(w), W(w), h(w)$

Summary

- We formulated a Two-Stage Stochastic Linear Programming model for the stochastic inventory control problem.
- We reformulated it into a deterministic linear program when uncertainty is a discrete random variable.
- We introduced general two-stage stochastic programming model.